RodenZ: Robust Probabilistic Triangulation of Moving Mice

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ABSTRACT

Accurate three-dimensional tracking of movement from sequential video data is fundamental to the study of animal behavior, with a widespread approach being geometric triangulation on multi-camera sequences of 2D keypoints. These methods require accurate prior accurate camera calibration. In the context of missing calibration, the use of animal subject keypoints for joint optimization of camera parameters has been proposed. This introduces an overreliance on possibly faulty detections, and disregards the temporal structure inherent to the data. In this work, we propose leveraging tools of the Bayesian Smoothing domain to address these limitations. Our approach enables robust calibration and triangulation even in scenarios where reliable calibration sequences are unavailable, while exploiting temporal context for the inference of *all* parameters, positions and calibrations, within a unified, single-stage pipeline. We demonstrate the effectiveness of our method on a dataset of freely moving mice recorded with an experimental setup of modest quality, showing highly competitive performance on the 3D pose reconstruction task.

Code: https://github.com/thws-mai/PM25_Mice

1 INTRODUCTION AND RELATED WORK

The analysis of animal body movements provides critical insights across a wide range of fields, especially in neurology [20] and biology [1, 12]. A frequently used method involves extracting two-dimensional positions of skeletal keypoints from video data of animals. CNN-based feature extractors such as DeepLabCut (DLC) [15] and others[4, 6, 13, 17, 18] facilitate this tracking with only few labeled frames.

However, many complex or subtle tasks require accurate 3D pose estimates, which requires lifting these 2D observations into a 3D context. Existing tools for 3D pose estimation from sequences of 2D keypoints can generally be categorized into two main approaches:.

- **Geometric:** This approach leverages multi-view geometry, such as triangulation [16], or techniques like bundle adjustment (BA) [7].
- **Probabilistic**: These approaches model pose estimation as a probabilistic inference problem, using Bayesian filtering techniques such as the Extended Kalman Filter (EKF) [9].

Along with these approaches for reconstruction from 2D detections, there exists a third category of direct image-to-3D approaches based on deep learning [5, 22], which have shown good results. However, these methods are heavy on computing resources, and as a result most effectively applied to short videos. However, in this work we work with videos of around 15 minutes in length, and don't consider these methods further.

Traditionally, all of the named approaches depend on high-quality a priori camera calibration, relying on video sequences containing a checkerboard or other calibration pattern with known geometry to estimate the camera parameters [21]. However, in many real-world scenarios, board calibrations are not realistic to obtain, either because of miniscule experiment size [7], or because of operational issues, as is the case for our dataset. In such situations, the usage of Bundle Adjustment (BA), an optimization framework where relative camera geometry and locations of 2D observations in 3D are esimated, has been proposed to estimate camera geometry directly from the points obtained by the keypoint detector. [7, 10].

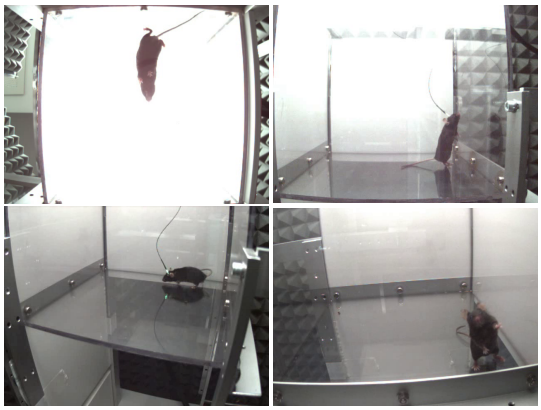

Figure 1: Example frames from each of the four cameras.

Although this approach can work in practice, it suffers from two fundamental drawbacks:

- **Overconfidence in Keypoint Detector.** Using outlier detections for BA might significantly hamper the calibration effort. This is usually addressed by only using keypoints the detector has flagged as being 'high-likelihood' [7, 10], which can only serve as a mitigation, since the detector might still produce confident false positives.
- **Disregarding Temporal Context.** Treating the detections as individual reference points for BA loses the temporal context of the detections being points on an animal's body in time. Integrating this temporal knowledge could make for a more data-efficient approach, and lead to more robustness to the mentioned outlier detections.

Addressing these limitations, we introduce RodenZ, a robust probabilistic triangulation method for tracking freely moving mice, based on an Unscented Kalman Filter (UKF). Whereas previous probabilistic approaches only integrated temporal context into animal positions [9], RodenZ directly estimates its extrinsic and intrinsic camera parameters within the state model, integrating temoral context into the calibration process, resulting in a more robust holistic approach to the combined calibration-tracking problem.

2 A CHALLENGING DATASET OF MICE

RodenZ was developed whilst working on a challenging dataset of mice recorded from four camera perspectives (Figure 1), the raw video data having been collected by a team at the University

Clinic of Würzburg. The focus of their research is the study of the impact of optogenetics on animal behaviour, a technique where neurons are stimulated using light in an attempt to elucidate their functionality[8, 14].

From the raw video data, we extracted 800 Frames, uniformly distributed across camera views, by using the k-means frame clustering approach from DeepLabCut [16], producing a set of frames showing mice in a variety of poses. These frames were manually annotated with the 2D image positions of 20 keypoints, including limbs, the spine and facial keypoints.

We applied a cross-validation-like approach in which, for each camera, a unique split of approximately 65 view-dependent frames was extracted for evaluation, while a CNN was trained on the remaining annotations. All extracted annotations originate from the same video, which was fully excluded from the training set to prevent data leakage. This strategy allows us to make the most of our limited data, as each annotation is ultimately used in the full evaluation. These annotations serve as the ground-truth 2D positions, which are then compared to the 2D projections of the 3D estimates to compute our quantitative error measure.

In terms of board calibration sequences, there is a sequence of ChArUco-calibrations for three of the four cameras. Unfortunately, the bottom camera (see Figure 1) was broken when the calibration sequences were recorded and was subsequently removed from the setup in order to be repaired, making it unlikely that a calibration sequence from the exact previous position of this camera will ever be recorded. In particular, this means that approaches building on checkerboard sequences are not able to use this camera.

Aside from the missing calibration sequence, there are other factors that make this dataset challenging.

For one, whilst many other datasets [9, 10] feature image resolutions of 1920×1080 or higher, the footage we were provided with sits at a very modest 640×480, with the actual mouse occupying a significantly smaller portion of the frame in most situations (see Figure 1). Intuitively, this constitutes a significant limiting factor to reconstruction fidelity.

Moreover, the actual camera geometry of the setup (Figure 1), is somewhat suboptimal. Unlike other experiments [7, 9], there is no camera "behind" the box, meaning that there are potential dead angles where key points are not visible from any camera.

In combination, the lacking calibration, poor camera geometry and modest resolution make this a far more challenging scenario than what is usually discussed in the literature, and will require a principled, data-efficient processing of all available knowledge over time to achieve good calibration and reconstruction.

3 METHODOLOGY

Using the training sets described in Section 2, we trained a DLC keypoint detector [16], and used it to annotate entire sets of synchronized videos.

Our method aims to compute a probability distribution of 3D mouse keypoint positions, camera locations and focal lengths at every timestep given the entire sequence of 2D keypoint detections produced by DLC for a given set of videos. We model these distributions as Gaussian and recursively calculate their parameters using a Bayesian smoothing approach [3, 19]. In the remainder of this section, we describe the modeling assumptions we made for deploying our approach. We expect the reader to be well aquainted with Bayesian Filtering [19] and fundamental computer vision topics such as the pinhole camera model and homogeneous coordinates [11].

3.1 State Model

We need to consider keypoints and camera positions in the state model. Intuitively, if we were to consider all camera positions as unknown, the system would be overdetermined, as our measurements only contain information about relative observations of the state. For this reason, we choose one of our cameras as an "anchor", fixing the camera position of that camera to some sensible default.

The states are then vectors $\mathbf{x} \in \mathbb{R}^{D_1}$. The state contains six parameters per keypoint, seven parameters per non-anchor camera, and a field of view angle for the anchor camera, so in total $D_1 = 6K + 7(C-1) + 1$, where K is the number of keypoints and C is the number of cameras. These parameters are arranged in a state vector as follows:

$$\begin{aligned}\mathbf{x} = ((\mathbf{x}'_1)^T, ..., (\mathbf{x}'_K)^T, (\dot{\mathbf{x}}'_1)^T, ..., (\dot{\mathbf{x}}'_K)^T, \gamma_1,\\ \Theta_2, \Phi_2, r_2, \theta_2^{(y)}, \theta_2^{(x)}, \theta_2^{(z)}, \gamma_2, ...\\ \Theta_C, \Phi_C, r_C, \theta_C^{(y)}, \theta_C^{(x)}, \theta_C^{(z)}, \gamma_C)^T\end{aligned}$$

Where $\mathbf{x}'_k, \dot{\mathbf{x}}'_k \in \mathbb{R}^3$ are positions and velocities of keypoint k, respectively. $\Theta_c, \Phi_c \in [0, 2\pi[$ and $r_c \in \mathbb{R}$ express the position of a 3D camera as altitude, azimuth and radius respectively. By default, a camera "looks" at the origin $\mathbf{0} \in \mathbb{R}^3$. To allow deviations in viewing direction, $\theta_c^{(y)}, \theta_c^{(x)}, \theta_c^{(z)} \in [0, 2\pi[$, forming yaw, pitch and roll angles respectively, make up for the remaining three degrees of freedom to produce a complete parametrization of camera extrinsics. $\gamma_c \in [0, \pi[$ serves to describe the camera's field of view, as the only parameter of camera intrinsics in our state. We assume measurements to be in normalized coordinates with any radial distortion already removed and aspect ratios $a_1, ..., a_C$ known, which saves us the trouble of tracking the remaining intrinsic parameters.

Observations $\mathbf{y} \in \mathbb{R}^{D_2}$, $D_2 = C \cdot K \cdot 2$, are then simply these normalized 2D positions $\mathbf{y'}_k^{(c)}$, of every keypoint k, seen from every camera c:

$$\mathbf{y} = ((\mathbf{y'}_1^{(1)})^T, ..., (\mathbf{y'}_K^{(1)})^T, ..., (\mathbf{y'}_K^{(C)})^T)^T$$

The goal is to estimate $p(\mathbf{x}_t | \mathbf{y}_1, ..., \mathbf{y}_T) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t, \mathbf{P}_t)$, for every frame $t \in \{1, ..., T\}$ of a video. We will discuss our specific choice of prior $\mathbf{m}_0, \mathbf{P}_0$ in Section 4.

3.2 State Transition Model

Being a simple first-order state model, we model the time-dependency between states as a linear function (adding the velocity to the position) plus a Gaussian random walk on the velocity component:

$$\mathbf{x}_{t+1} = \begin{bmatrix} \mathbf{I}^{(3K\times 3K)} & \Delta_t \mathbf{I}^{(3K\times 3K)} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \mathbf{x_t} + \begin{pmatrix} \mathbf{0}^{(3K)} \\ {}^x\boldsymbol{\xi}_t \\ \mathbf{0}^{(7(C-1)+1)} \end{pmatrix}$$

$${}^x\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \Delta_t^2 \sigma_v \mathbf{I})$$

Where Δ_t is the time interval between frames t and $t+1$, and σ_v is the variance of the random walk on the velocity component, chosen as a hyperparameter by the user.

3.3 Measurement Model

The measurement model represents our belief of how the states $\mathbf{x}_t$ "become" the measurements $\mathbf{y}_t$. Algorithm 1 implements the deterministic part of this procedure. In words, the algorithm first forms

Algorithm 1 Deterministic State Measurement $\mathbf{h}(\mathbf{x}_t)$

Require: $\mathbf{x}_t, \Theta_1, \Phi_1, r_1, \alpha_1^{(y)}, \alpha_1^{(x)}, \alpha_1^{(z)}, \gamma_1, a_1, ..., a_C$
  $\tilde{\mathbf{y}}_t \leftarrow \mathbf{0}$
  for $c = 1, \ldots, C$ **do**
    $\mathbf{V}_c \leftarrow \mathbf{R}_z(\theta_c^{(z)}) \cdot \mathbf{R}_x(\theta_c^{(x)}) \cdot \mathbf{R}_y(\theta_c^{(y)}) \cdot \mathbf{T}_z(-r_c) \cdot \mathbf{R}_x(\Theta_c) \cdot \mathbf{R}_y(\Phi_c)$
    $f_c \leftarrow tan(\frac{1}{2}\gamma_c)^{-1}$
    $\mathbf{P}_c \leftarrow \begin{bmatrix} \frac{f_c}{a_c} & 0 & 0 & 0 \\ 0 & f_c & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$
    for $k = 1, \ldots, K$ **do**
      $\mathbf{p} \leftarrow \mathbf{P}_c \mathbf{V_c} \begin{pmatrix} \mathbf{x'}_k \\ 1 \end{pmatrix}$
      $\tilde{\mathbf{y'}}_k^{(c)} \leftarrow \frac{1}{\mathbf{p}_3}(\mathbf{p}_1 \ \mathbf{p}_2)^T$
    end for
  end for
  $\mathbf{h}(\mathbf{x}_t) \leftarrow \tilde{\mathbf{y}}_t$

view matrices $\mathbf{V}_c$ and projection matrices $\mathbf{P}_c$ for every camera. The view matrices are formed by multiplying a series of homogenous rotation matrices $\mathbf{R}_{[\,\cdot\,]}(\cdot)$, where the subscript denotes axis of rotation, and homogenous translation matrices $\mathbf{T}_{[\,\cdot\,]}(\cdot)$, where the subscript denotes the axis of translation. The projection matrices are formed from the field of view parameters γ_c and aspect ratios a_c. Finally, we simply project each keypoint into each camera using these matrices.

To account for noise in the measurement process, we consider our measurement model to be this deterministic projection plus some additive Gaussian noise in image space:

$$\mathbf{y}_t = \mathbf{h}(\mathbf{x}_t) + {}^y\boldsymbol{\xi}_t$$
$${}^y\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, diag({}^y\boldsymbol{\sigma}_t))$$

Where the covariances we consider for our measurements ${}^y\boldsymbol{\sigma}_t$ depend on how much we "trust" our keypoint detector at time t. In practice, when the keypoint detector reports a high uncertainty for a given keypoint k in a given camera c, we will increase the respective variance entries in ${}^y\boldsymbol{\sigma}_t$, whereas we will set some lower value for those keypoints where the detector is "sure" about it's prediction.

We then obtain the parameters $\mathbf{m}_t, \mathbf{P}_t$ for the distributions $p(\mathbf{x}_t|\mathbf{y}_{1:T}) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t, \mathbf{P}_t)$ by running a forward pass of Gaussian filtering (UKF) using these three models, and a backward pass of RTS-Smoothing [19].

Our methodology is implemented as a collection of command-line tools, each guiding the user through one step of the pipeline (inference of 2D keypoints, removal of radial distortion, and finally reconstruction), and may be found using the GitHub link in the abstract.

4 RESULTS

We compared RodenZ to the predominant BA-Triangulation methodolgy, calculating the reprojection error of inferred 3D sequences to 2D labels from the *evaluation set* of Section 2 as a quantitative score.

For both methodologies, we first trained a DLC keypoint detector using the *training set* described in Section 2, and inferred 2D keypoint positions for the videos from which we extracted the labeled frames of the *evaluation set*. We then used RodenZ and AniPose to reconstruct 3D trajectories and camera poses for these videos.

For RodenZ, we had to decide on a prior distribution of 3D positions and camera poses, measurement variances ${}^y\boldsymbol{\sigma}_t$ and a velocity variance σ_v.

As the real mouse box is a cube of roughly $30cm$ side length, we chose the imagined origin of our frame of reference to be the center of the cube floor, and used cm as the units to describe the distance-related hyperparameters. We then chose camera positions of the format described in Section 3, by aligning a rendered virtual rendition of the box with the frames that can be seen in Figure 1, using the camera parameters obtained this way for the fixed anchor camera and prior means of the varying cameras respectively. For the mean prior mouse position, we assumed Gaussians centered at the floor of the box, with some prior variance covering the volume of the box generously.

For the measurement variances ${}^y\boldsymbol{\sigma}_t$, we assumed variance values based on thresholds set on the keypoint likelihood reported by the DLC predictor [16]. We used likelihood thresholds of 0.6 and 0.8. We observed empirically the two-dimensional variance of the difference of the training set annotations and the DLC outputs at these likelihood thresholds, and used these to set the values in ${}^y\boldsymbol{\sigma}_t$.

Choosing σ_v proved to be difficult to do in a principled way, as expert knowledge on "mouse agility" was not available at implementation time. In practice, we reconstructed some 3D trajectories (that were not part of the dataset), and simply observed the 3D reconstructions obtained for different values. We then chose the actual quantity by human judgment, based on if details of movement were visually preserved without observing the 'jumpiness' indicative of the RTS explaining outliers with high velocity noise. We settled on a variance of $\sigma_v = 5$ for our setup.

4.1 Evaluation

To facilitate a quantitative comparison, we picked out the reconstructed 3D positions at the timesteps where we have manual 2D labels in the evaluation set, and reprojected those 3D points to the image plane. The difference in reprojected 3D position and 2D ground truth labels was then used to estimate the quality of the 3D point. We used the distribution of 2D reconstruction error as our main method for comparison, keeping in line with the established literature [10].

For the BA-Triangulation methodology, we adopted the AniPose[10] package. As we have no calibration sequences for the bottom camera, the classical approach of calibrating with boards using bundle

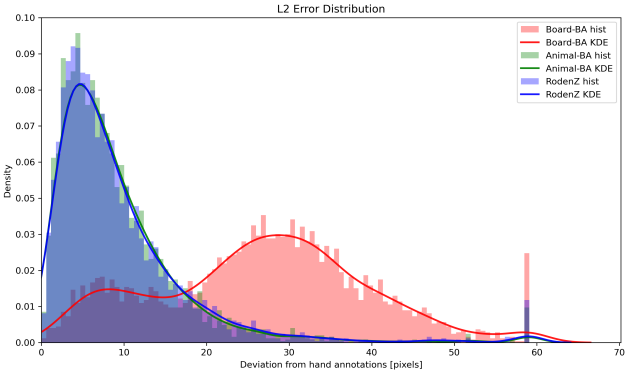

Figure 2: Error distribution for the three-camera task.

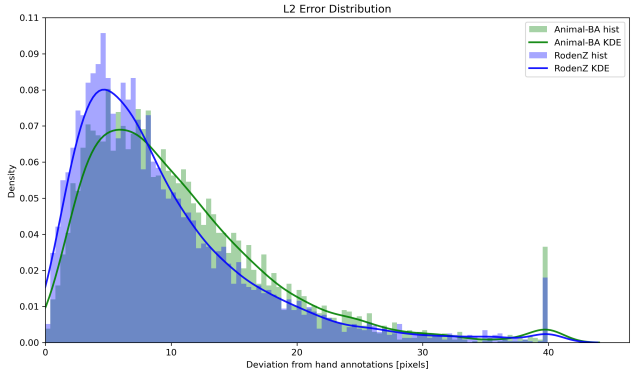

Figure 3: Error distribution for the full task.

adjustment (BA) and then triangulating, which we will call the *BA-Board* approach, is not deployable to the full setup in our case. Nonetheless, we ran this approach on the remaining three cameras, comparing it to a rendition of RodenZ artificially limited to the same three cameras. We also used a feature of AniPose, where the initial board calibrations can be further refined using bundle adjustment of detected keypoints [10], which we refer to as *Animal-BA*.

For the full setup of four cameras, we evaluated if extracting the camera calibrations learned by RodenZ and using them as starting parameters for a four-camera version of Animal-BA could perhaps lead to better reconstruction performance. This implements the established calibration from keypoints procedure described in Section 1, which requires plausible starting parameters [7]. In this way, we achieve a comparison of how good the calibration is on the four-camera setup: If BA found better calibrations, it would manifest as a decrease in reprojection error.

The distribution of error of each approach on the limited three-camera version of the dataset can be seen in Figure 2. We can see that in this three camera version of the dataset, the distribution of reprojection error of RodenZ and Animal-BA match almost exactly, both being vastly superior to the board calibrations alone. RodenZ does not exceed the classical performance on this task, it is nonetheless worthwhile to note that we can reproduce a very good result on this task without relying on initial board calibrations.

Of the approaches discussed here, only RodenZ is natively deployable to the four-camera setup. Observing the distribution of reconstruction errors for the full setup (Figure 3), we see RodenZ taking the lead over the Animal-BA version that was initialized with calibrations extracted from the RodenZ run. Animal-BA noticably degrades the reconstruction error, even though it starts from initial calibrations of very high quality, as indiciated by the good RodenZ performance. A potential reason for the degrading of calibrations through Animal-BA in this case might lie with the camera perspective of the added camera being unlike the others. As the previous three perspectives were all taken from the side, there was less training data in the training set for the 'bottom' perspective, possibly leading to worse keypoint detector performance in that camera. This may have enabled confident false positive keypoints, leading the BA optimization away from the good initial calibrations.

Overall, the quantitive experiments display some nuanced results. Whilst we did not manage to beat the established methodology of Animal-BA for the three-camera subtask, underscoring the effectiveness of the established method, we observed superior performance in reprojection when setting out from calibrations supplied by our approach for the four-camera task. We attribute this to our approach being more robust to outliers that result from overconfident keypoint detections in an unusual camera perspective.

5 LIMITATIONS

We only measured the reprojection error for a quantitive evaluation of our approach, which summarizes over many aspects of a complex reconstruction system. One important quantity that we did not have the opportunity yet to evaluate is fitness for medical or biological processing down the line. A potential threat to the validity of our approach is that important aspects of animal behavior that only manifest in very fine movements are simply not visible in the smoothing output, because of the increased filtering of outliers.

Should it be the case that the 3D reconstructions are unfit for downstream processing, one could still use RodenZ as a standalone calibration procedure, and use the parameters obtained this way in a triangulation scheme that keeps these important details better preserved.

6 CONCLUSION AND FUTURE WORK

We demonstrated a robust probabilistic triangulation and calibration approach on a challenging task of skeletal keypoint reconstruction. By designing a unified pipeline, we were even able to integrate information from non-calibrated cameras. Furthermore, our approach matches SotA on the calibratable subset of our task, and outperforms competing methods on the full task, even when competing approaches are allowed to exploit the calibration parameters obtained from our approach.

In the future, our approach could be refined by leveraging recent research on improved keypoint detectors [2], or be extended to include anatomical constraints in the state transition model, such as joint lengths [9]. Also, one could strive to include radial camera distortion parameters in the state estimation, if these are not easy to obtain a priori for a given scenario.

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